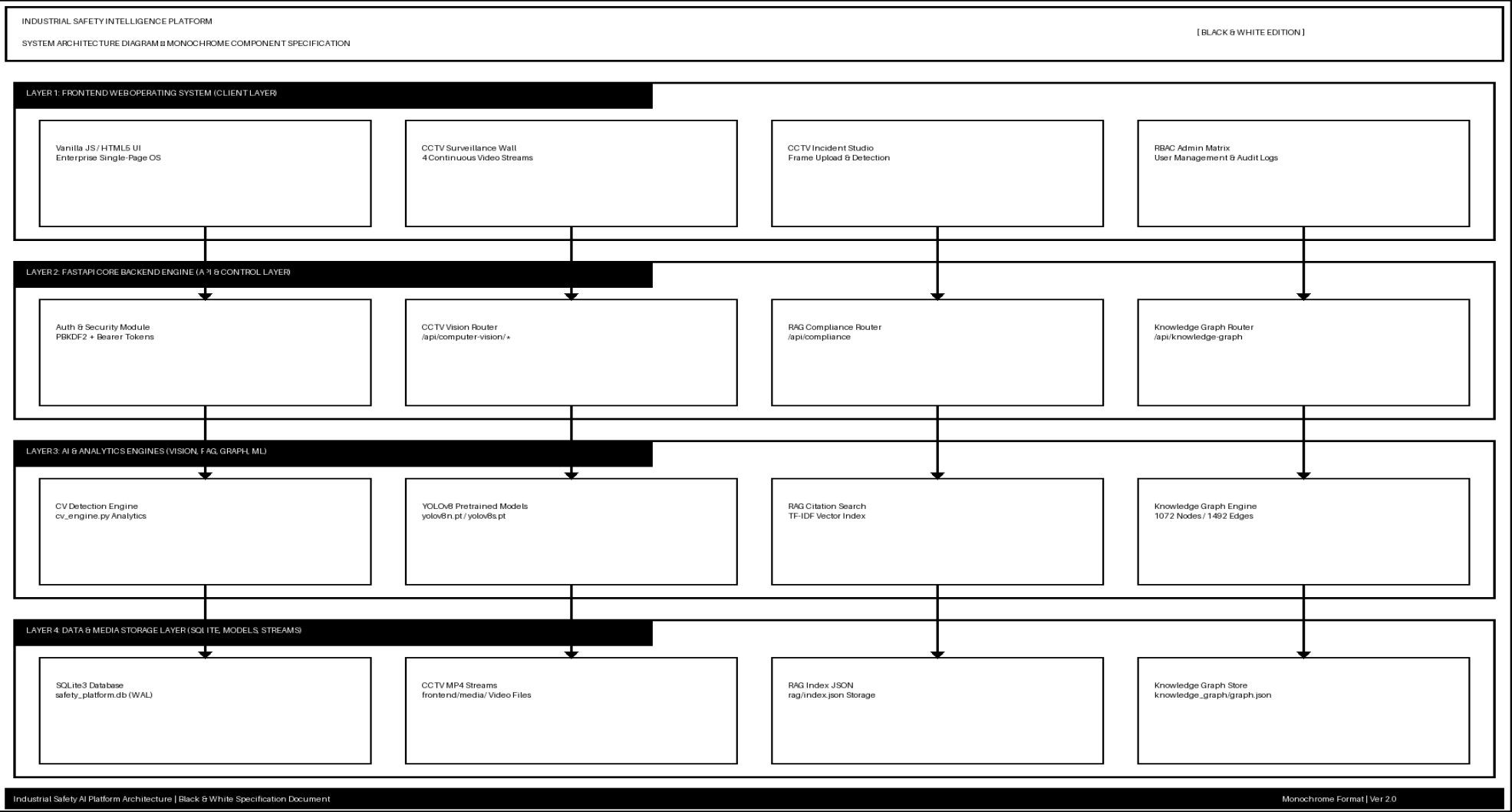


Industrial Safety Intelligence Platform — System Architecture

Monochrome Specification & Multi-Layer System Diagram (Black & White Edition)



Layer Component	Technology Stack	Architecture Description / Specification
Client Interface	Vanilla HTML5 / CSS3 / ES6 JS	4-Channel CCTV Continuous Live Feed Wall & Incident Studio
API Backend Engine	FastAPI + Uvicorn (Python 3.13)	Async REST APIs, Authentication, RBAC Matrix, Audit Logging
AI & CV Core	YOLOv8 (yolov8n.pt / yolov8s.pt) + PyTorch	PPE Violation, Fallen Worker (Man Down), Vehicle Proximity, Fire/Smoke

RAG & Knowledge Graph	Vector Index + NetworkX Graph	Grounded OSHA/NIST Regulatory Citations & 1072-Node Risk Graph
Data & Storage	SQLite3 (WAL Mode) + Media Store	Relational Database, Models Registry, CCTV Streams, Snapshots